Grade 11 – Term 1
C1–C7 Catch-up Exam Solutions
Continuous distributions and sample data

Global Economics – 2026–2027

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Learning objective

Identify probability distributions and their supports; use geometric interval probabilities and exponential waiting times; calculate theoretical moments, empirical probabilities, and sample statistics; and interpret errors relative to theoretical models in economic and operational contexts.

Evaluated criteria

C1. Characterises discrete and continuous random variables, identifies suitable distributions and supports, and interprets density and cumulative probability in context.

C2. Applies uniform and triangular models and constant or piecewise-linear densities to interval probabilities using geometric areas.

C3. Uses circular geometry and arc proportions to calculate probabilities for uniform timing and directions around a circle.

C4. Applies exponential models to constant-rate waiting times, infers event rates, and assesses model suitability.

C5. Calculates and interprets the mean, variance, and standard deviation of continuous distributions using standard formulas.

C6. Estimates interval probabilities using sample relative frequencies and compares them with theoretical probabilities.

C7. Calculates sample means, variances, and standard deviations for different sample sizes and data representations and compares them with theory.

Assessment information

Each task shows its points. Passing requires **at least 5 cumulative points per criterion, including previous learning activities**. Show all procedures; answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Missing names or late submissions incur a 1-point deduction per criterion.

Criterion	C1	C2	C3	C4	C5	C6	C7	Total
Points	6	6	6	7	6	6	6	43

Teacher note. The source assessment table lists Problem 3 as 2 points although its heading and tasks award 3. We follow the headings and tasks: C3 totals 6 points. The source table also lists Problem 6 as 3, but its heading and tasks award 2; C4 therefore totals 7. The source exam has not been changed.

Conventions. $N(\mu, \sigma^2)$ uses mean and variance; $\mathrm{Tri}(a, b, c)$ means minimum, maximum and mode; $\mathrm{Exp}(\lambda)$ uses the rate parameter. Supports include their finite boundary points; endpoint inclusion does not affect continuous probabilities. Probabilities are rounded to four decimal places (percentages to two), statistics, rates and times to three where needed, and percentage errors to two. All errors use unrounded results. Sample variance uses $n - 1$. Blue task text links to its solution; each solution links back to its task. The arrows return to the problem and contents.

1 Models in organisations [C1-6]

Colombian organisations model two random variables in each context below. **For each variable**: classify it as discrete or continuous, and identify its distribution.

Questions/Tasks

C1 1. Cooperative repayments. A cooperative records (a) whether one randomly selected member repays a monthly instalment on time, coded as 1 if timely and 0 otherwise; and (b) the number of timely repayments among 40 members. Each member repays on time independently with probability 0.92. **[1]**

C1 2. Warehouse quality checks. A warehouse records (a) the number of invoices checked until the first incorrect invoice, including that invoice, with independent checks and a constant error probability of 0.08; and (b) a scale's measurement error between -3 and 3 grams, with equal-length intervals having equal probabilities and no errors outside these limits. **[1]**

C1 3. Printing orders. A printer records (a) the number of orders received in one hour, arriving independently at a constant rate of 2 per hour, one at a time, with multiple arrivals in a sufficiently short interval negligible; and (b) a poster order's completion time on $[10, 30]$ minutes, whose density rises linearly from zero at 10 to a single peak at 18, then falls linearly to zero at 30, with no values outside this range. **[1]**

C1 4. Coffee production. A roaster records (a) the fraction of a storage tank used at dispatch on $[0, 1]$, whose density increases along one straight line from zero at 0 to its maximum at 1 and is zero elsewhere; and (b) a filling machine's measurement error, modelled by an exact symmetric bell curve on the real line, centred at 0 grams with standard deviation 4 grams. **[1]**

C1 5. Digital shop operations. A digital shop records (a) the number of refund requests among 50 purchases, each independently resulting in a request with probability 0.04; and (b) the number of customer enquiries received over three days, arriving independently at a constant rate of 5 per day, one at a time, with multiple arrivals in a sufficiently short interval negligible. **[1]**

C1 6. Urban delivery services. A delivery company in Bogotá studies two aspects of its daily operations. (a) Delivery times range from 15 to 45 minutes. Short deliveries are the most common, and the likelihood of observing longer delivery times decreases steadily until reaching zero at the upper limit. (b) The company also monitors the weight of its packages. Most packages weigh close to 5 kg, while lighter and heavier packages become progressively less common in a symmetric pattern. The standard deviation of package weights is 1.2 kg. **[1]**

Solution: Cooperative repayments

(a) Discrete: $X \sim \text{Bernoulli}(0.92)$, support $\{0, 1\}$. One member gives a single timely/not-timely outcome. **(b)** Discrete: $Y \sim \text{Bin}(40, 0.92)$, support $\{0, 1, \ldots, 40\}$. This counts successes in 40 independent trials with the same success probability.

Return to this task

Solution: Warehouse quality checks

(a) Discrete: $X \sim \text{Geom}(0.08)$, support $\{1, 2, \ldots\}$. We count all trials up to and including the first incorrect invoice; $P(X = k) = 0.92^{k-1}(0.08)$. **(b)** Continuous: $Y \sim U(-3, 3)$ grams, support $[-3, 3]$ g. Equal-length intervals have equal probabilities, so $f(y) = 1/6$ on the support, zero elsewhere.

Return to this task

Solution: Printing orders

(a) Discrete: $X \sim \text{Poisson}(2)$, support $\{0, 1, 2, \ldots\}$. The one-hour mean is $2 \times 1 = 2$ orders and the stated arrival conditions define a Poisson process. **(b)** Continuous: $Y \sim \text{Tri}(10, 30, 18)$ minutes, support $[10, 30]$ min. The two straight branches and single peak specify the triangular model with mode 18.

Return to this task

Solution: Coffee production

(a) Continuous: increasing-linear density on $[0, 1]$, equivalently $X \sim \text{Tri}(0, 1, 1)$. Normalisation gives $f(x) = 2x$ on $[0, 1]$, zero elsewhere. The fraction is unitless, and the endpoint mode is 1. **(b)** Continuous: $Y \sim N(0, 16)$ in grams, support $\mathbb{R}$. The exact symmetric bell curve identifies the normal model, with $\sigma = 4$ g and variance $4^2 = 16$ g^2.

Return to this task

Solution: Digital shop operations

(a) Discrete: $X \sim \text{Bin}(50, 0.04)$, support $\{0, 1, \ldots, 50\}$. There are 50 independent purchases with the same refund probability. **(b)** Discrete: $Y \sim \text{Poisson}(15)$, support $\{0, 1, 2, \ldots\}$. The three-day mean is $5 \times 3 = 15$ enquiries, rather than 5.

Return to this task

Solution: Urban delivery services

(a) Continuous, support $[15, 45]$ min. "Decreases steadily" does not uniquely establish a straight-line density. *If a decreasing straight line is intended*, use $X \sim \text{Tri}(15, 45, 15)$ with $f(x) = (45 - x)/450$ on this support and zero elsewhere. Its height is $2/30 = 1/15$ min^{-1}; no extra free parameter is required under this assumption. **(b)** Continuous package mass. Symmetric concentration around 5 kg and SD 1.2 kg do not uniquely determine a normal distribution or its full support. *If a normal shape is assumed*, $Y \sim N(5, 1.44)$ in kg, with model support $\mathbb{R}$ and variance $1.2^2 = 1.44$ kg^2. Actual package masses are nonnegative, but no exact physical support is supplied. These are intended candidates under explicit assumptions, not uniquely implied distributions.

Return to this task

2 Market-stall setup [C2-6]

The setup time S for a stall at a Bogotá craft market ranges from 5 to 25 minutes. Its density is zero at both endpoints, rises linearly to its single maximum at 13 minutes, then falls linearly to zero; it is zero outside $[5, 25]$.

Questions/Tasks

C2 Find $P(9 \le S \le 19)$. **[3]**

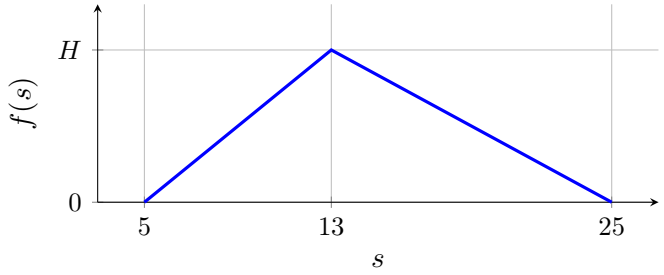
C2 Suppose the range changes to $[0, 30]$, with other properties unchanged. Recalculate $P(9 \le S \le 19)$. **[3]**

Solution: Original range

Stage 1: Support and shape

$S \sim \text{Tri}(5, 25, 13)$ has support $[5, 25]$ min. The density rises from $(5, 0)$ to $(13, H)$ and falls to $(25, 0)$.

Support and triangular shape; horizontal coordinate in minutes



Stage 2: Total area = 1

The triangle area is $\frac{1}{2} \times \text{base} \times \text{height}$, hence

$$1 = \frac{1}{2}(25 - 5)H \quad \Longrightarrow \quad H = \frac{2}{20} = \frac{1}{10} \text{ min}^{-1}.$$

Stage 3: Complete piecewise pdf

For each line, use $m = (y_2 - y_1)/(x_2 - x_1)$ and $q = y_1 - mx_1$ in $f(s) = ms + q$. The rising branch passes through $(5, 0)$ and $(13, \frac{1}{10})$:

$$m_1 = \frac{\frac{1}{10} - 0}{13 - 5} = \frac{1}{80}, \qquad q_1 = 0 - \frac{5}{80}, \qquad f_1(s) = \frac{s - 5}{80}.$$

The falling branch passes through $(13, \frac{1}{10})$ and $(25, 0)$:

$$m_2 = \frac{0 - \frac{1}{10}}{25 - 13} = -\frac{1}{120}, \qquad q_2 = 0 - \left(-\frac{1}{120}\right) 25 = \frac{25}{120}.$$

Therefore the complete density is

$$f(s) = \begin{cases} \dfrac{s - 5}{80}, & 5 \le s \le 13, \\ \dfrac{25 - s}{120}, & 13 < s \le 25, \\ 0, & \text{otherwise.} \end{cases}$$

Both branches meet at height $\frac{1}{10}$, are nonnegative, and have total area 1.

Stage 4: Probability by complementary tail triangles

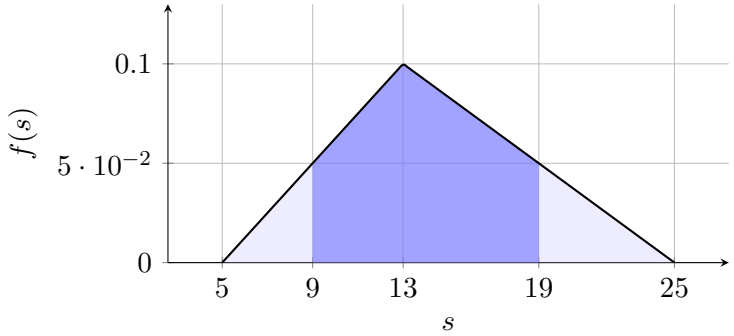
The event straddles the mode. Remove the left triangle up to 9 and the right triangle after 19:

$$P(9 \le S \le 19) = 1 - \frac{1}{2}(9-5)f(9) - \frac{1}{2}(25-19)f(19).$$

Here $f(9) = 4/80 = 1/20$ and $f(19) = 6/120 = 1/20$ min^{-1}, so

$$P = 1 - \tfrac{1}{2}(4)(1/20) - \tfrac{1}{2}(6)(1/20) = 1 - \frac{1}{10} - \frac{3}{20} = \boxed{\frac{3}{4} = 0.7500 = 75.00\%}.$$

Shaded event: $9 \le S \le 19$; s in minutes



Density units are inverse minutes. Integrating the two branches over $[9, 13]$ and $[13, 19]$ gives the same result.

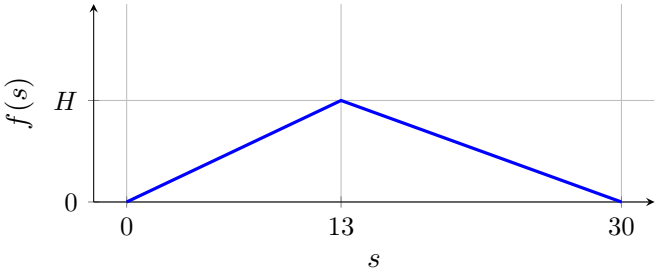
Return to this task

Solution: Changed range

Stage 1: Support and shape

Interpretation. "Other properties unchanged" retains the triangular shape and the mode at 13 minutes. Only the endpoints change to 0 and 30; the mode is not moved to 15. $S \sim \text{Tri}(0, 30, 13)$ has support $[0, 30]$ min. The density rises from $(0, 0)$ to $(13, H)$ and falls to $(30, 0)$.

Support and triangular shape; horizontal coordinate in minutes



Stage 2: Total area = 1

The triangle area is $\frac{1}{2} \times$ base $\times$ height, hence

$$1 = \frac{1}{2}(30-0)H \quad \Longrightarrow \quad H = \frac{2}{30} = \frac{1}{15} \text{ min}^{-1}.$$

Stage 3: Complete piecewise pdf

For each line, use $m = (y_2 - y_1)/(x_2 - x_1)$ and $q = y_1 - mx_1$ in $f(s) = ms + q$. The rising branch passes through $(0, 0)$ and $(13, \frac{1}{15})$:

$$m_1 = \frac{\frac{1}{15} - 0}{13 - 0} = \frac{1}{195}, \qquad q_1 = 0 - \frac{0}{195}, \qquad f_1(s) = \frac{s - 0}{195}.$$

The falling branch passes through $(13, \frac{1}{15})$ and $(30, 0)$:

$$m_2 = \frac{0 - \frac{1}{15}}{30 - 13} = -\frac{1}{255}, \qquad q_2 = 0 - \left(-\frac{1}{255}\right) 30 = \frac{30}{255}.$$

Therefore the complete density is

$$f(s) = \begin{cases} \dfrac{s - 0}{195}, & 0 \le s \le 13, \\ \dfrac{30 - s}{255}, & 13 < s \le 30, \\ 0, & \text{otherwise.} \end{cases}$$

Both branches meet at height $\frac{1}{15}$, are nonnegative, and have total area 1.

Stage 4: Probability by complementary tail triangles

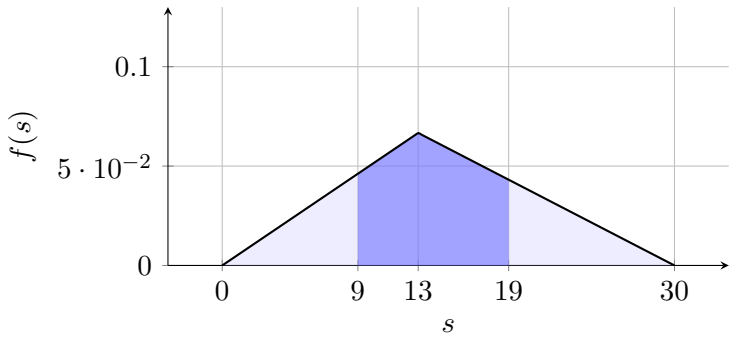
The event straddles the mode. Remove the left triangle up to 9 and the right triangle after 19:

$$P(9 \le S \le 19) = 1 - \frac{1}{2}(9 - 0)f(9) - \frac{1}{2}(30 - 19)f(19).$$

Here $f(9) = 9/195 = 3/65$ and $f(19) = 11/255$ min^{-1}, so

$$\begin{aligned} P &= 1 - \tfrac{1}{2}(9)(3/65) - \tfrac{1}{2}(11)(11/255) \\ &= 1 - \frac{27}{130} - \frac{121}{510} = \boxed{\frac{368}{663} \approx 0.5551 = 55.51\%}. \end{aligned}$$

Shaded event: $9 \le S \le 19$; s in minutes



Density units are inverse minutes. Integrating the two branches over $[9, 13]$ and $[13, 19]$ gives the same result.

Return to this task

3 A repeating inspection cycle [C3-3]

A cold-chain alarm occurs at a uniformly distributed position within a repeating 60-minute inspection cycle. The cycle starts at minute 0 and restarts immediately after minute 60.

Questions/Tasks

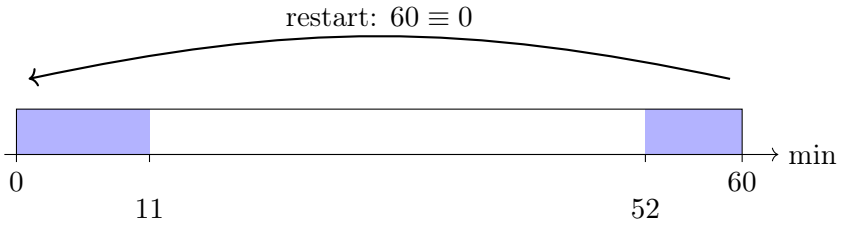
C3 Find the probability that the alarm occurs in the window from minute 52 through the restart to minute 11 of the next cycle. **[1]**

C3 Find the probability that the alarm occurs within 200 seconds before or after the start of a cycle. **[2]**

Solution: Window through the restart

Let $U \sim U(0, 60)$ represent the position in minutes. Its density is $1/60$ min^{-1} on $[0, 60)$. The restart splits the window into $[52, 60)$ and $[0, 11]$, with lengths 8 and 11 minutes.

The shaded lengths meet through the restart



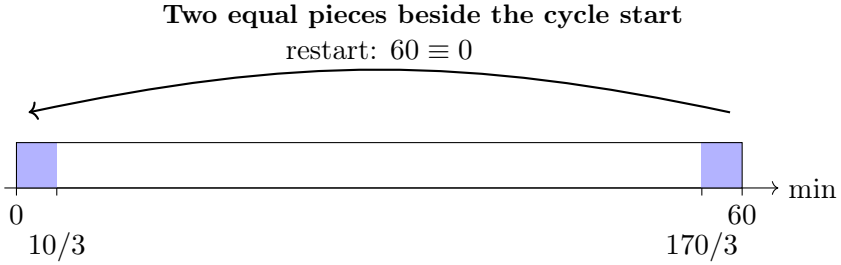
$$P = \frac{(60 - 52) + (11 - 0)}{60} = \frac{19}{60} \approx 0.3167 = 31.67\% \,.$$

Return to this task

Solution: Tolerance before and after a cycle start

Convert units first: 200 s $= 200/60 = 10/3$ min on *each* side. The intervals are $[0, 10/3]$ and $[60 - 10/3, 60) = [170/3, 60)$.

Two equal pieces beside the cycle start



$$P = \frac{10/3 + 10/3}{60} = \frac{1}{9} \approx 0.1111 = 11.11\% \,.$$

The endpoints have probability zero, so their inclusion does not change either answer.

Return to this task

4 Direction of a survey drone [C3-3]

A survey drone's initial heading is uniformly distributed around a full circle. A reference marker points north. "Within an angle" means that tolerance on **either side** of north, including wraparound.

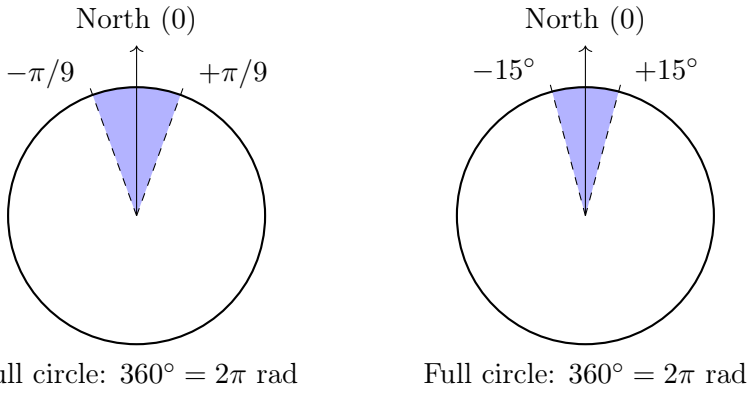
Questions/Tasks

C3 Compare the probability of a heading within $\pi/9$ radians of north with that of a heading within $15°$ of north. Calculate both probabilities and state which exceeds the other and by how much. **[3]**

Solution: Both angular probabilities and their comparison

Uniform directions give probability equal to favourable angle divided by full-circle angle. The tolerance is measured on *both sides* of north.

Two wraparound sectors centred at north



In radians, the event consists of $[0, \pi/9]$ and $[2\pi - \pi/9, 2\pi)$:

$$P_{\pi/9} = \frac{2(\pi/9)}{2\pi} = \boxed{\frac{1}{9} \approx 0.1111 = 11.11\%}.$$

In degrees, it consists of $[0, 15]$ and $[345, 360)$:

$$P_{15°} = \frac{2(15°)}{360°} = \boxed{\frac{1}{12} \approx 0.0833 = 8.33\%}.$$

Thus the $\pi/9$ tolerance (20 degrees on either side) has greater probability:

$$\Delta P = \frac{1}{9} - \frac{1}{12} = \boxed{\frac{1}{36} \approx 0.0278},$$

an absolute difference of **2.78 percentage points**. The relative percentage increase, using the 15-degree probability as the baseline, is

$$100\frac{1/36}{1/12}\% = 33.33\%.$$

Return to this task

5 Charging-space availability [C4-3]

Charging spaces at a logistics centre become free according to a Poisson process. The opening rate differs between the four scenarios below but remains constant within each scenario. In each case, observe from a fixed time and let T be the waiting time in minutes for the next opening.

Questions/Tasks

C4 With an average of 5 openings every 30 minutes, find the probability of waiting less than 4 minutes, $P(T < 4)$. **[0.5]**

C4 With a mean waiting time of 4 minutes between openings, find the probability of waiting more than 6 minutes, $P(T > 6)$. **[0.5]**

C4 With an average of 3 openings every 20 minutes, find the probability of waiting between 2 and 7 minutes, $P(2 < T < 7)$. **[1]**

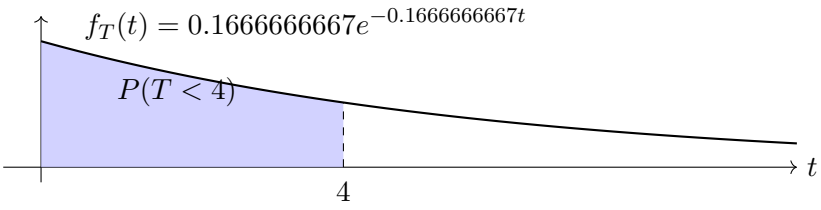
C4 With an average of 12 openings per hour, find the probability that the waiting time is outside the range from 3 to 8 minutes, $P(T < 3 \text{ or } T > 8)$. **[1]**

Solution: Less than four minutes

The rate is $\lambda_1 = 5/30 = 1/6 \approx 0.167$ openings per minute. For $T \sim \text{Exp}(\lambda_1)$, $P(T < t) = 1 - e^{-\lambda_1 t}$.

$$P(T < 4) = 1 - e^{-(1/6)(4)} = 1 - e^{-2/3} \approx 0.4866 = 48.66\% .$$

Shaded left tail; t in minutes and $f_T(t)$ in inverse minutes



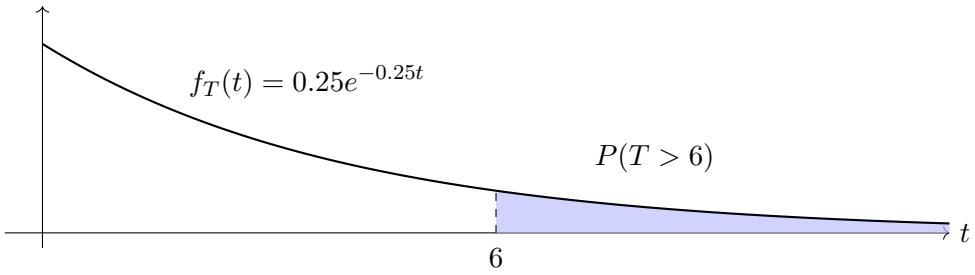
Return to this task

Solution: More than six minutes

A mean wait of 4 min gives $\lambda_2 = 1/E(T) = 1/4 = 0.250$ openings per minute. The survival formula is $P(T > t) = e^{-\lambda_2 t}$.

$$P(T > 6) = e^{-(1/4)(6)} = \boxed{e^{-3/2} \approx 0.2231 = 22.31\%}.$$

Shaded right tail; t in minutes and $f_T(t)$ in inverse minutes



The shaded tail continues beyond the plotted right edge to infinity.

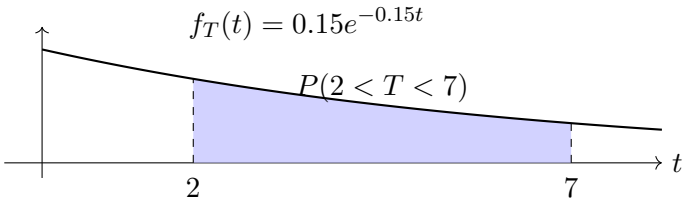
Return to this task

Solution: Between two and seven minutes

The separate rate is $\lambda_3 = 3/20 = 0.150$ openings per minute. For an interval, $P(a < T < b) = e^{-\lambda_3 a} - e^{-\lambda_3 b}$.

$$P(2 < T < 7) = e^{-(3/20)(2)} - e^{-(3/20)(7)} = \boxed{e^{-3/10} - e^{-21/20} \approx 0.3909 = 39.09\%}.$$

Shaded middle interval; t in minutes and $f_T(t)$ in inverse minutes



Return to this task

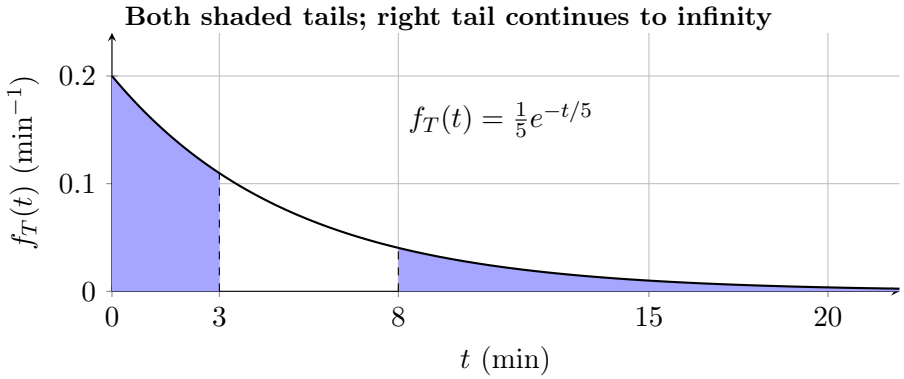
Solution: Outside three to eight minutes

Convert the hourly rate: $\lambda_4 = 12/60 = 1/5 = 0.200$ openings per minute. The tails are disjoint, so

$$P(T < a \text{ or } T > b) = (1 - e^{-\lambda_4 a}) + e^{-\lambda_4 b}.$$

$$P(T < 3 \text{ or } T > 8) = 1 - e^{-(1/5)(3)} + e^{-(1/5)(8)}$$

$$= \boxed{1 - e^{-3/5} + e^{-8/5} \approx 0.6531 = 65.31\%}.$$



For all four scenarios, $f_T(t) = \lambda e^{-\lambda t}$ for $t \geq 0$ and zero otherwise; support is $[0, \infty)$. The stated constant-rate Poisson process makes the exponential waiting-time model appropriate.

Return to this task

6 Electronic receipt rates [C4-2]

A digital shop issues electronic receipts according to a Poisson process with a constant rate during each shift. The morning and afternoon shifts have different rates. From a fixed observation time within each shift, let T_m and T_a be the waiting times in minutes for the next receipt in the morning and afternoon, respectively. Assume the relevant shift continues throughout each waiting time.

Questions/Tasks

C4 In the morning, $P(T_m > 10) = 0.40$. Infer the morning rate in receipts per hour. **[1]**

C4 In the afternoon, $P(T_a < 6) = 0.80$. Infer the afternoon rate in receipts per hour. **[1]**

Solution: Morning rate from survival

Use $P(T_m > t) = e^{-\lambda_m t}$:

$$0.40 = e^{-10\lambda_m} \implies \ln(0.40) = -10\lambda_m \implies \lambda_m = -\frac{\ln(0.40)}{10} \approx 0.092 \text{ receipts/min}.$$

Convert minutes to hours using the exact rate:

$$60\lambda_m = -6\ln(0.40) \approx \boxed{5.498 \text{ receipts/hour}}.$$

Return to this task

Solution: Afternoon rate from a cumulative probability

Use $P(T_a < t) = 1 - e^{-\lambda_a t}$:

$$0.80 = 1 - e^{-6\lambda_a} \implies e^{-6\lambda_a} = 0.20 \implies -6\lambda_a = \ln(0.20).$$

$$\lambda_a = -\frac{\ln(0.20)}{6} \approx 0.268 \text{ receipts/min}, \qquad 60\lambda_a = -10\ln(0.20) \approx \boxed{16.094 \text{ receipts/hour}}.$$

Return to this task

7 Two payment channels [C4-2]

Completed payments on channels A and B follow independent constant-rate Poisson processes. At a fixed observation time, the probability that A's next payment occurs within 5 minutes is 0.75. The probability that B's next payment takes more than 5 minutes is 0.50. Count payments on both channels from that same observation time.

Questions/Tasks

C4 Infer both rates. After how many minutes does A's expected payment count exceed B's expected count by 3? Find the time for this difference in expectations. **[2]**

Solution: Rates and the difference in expected counts

For A, the given probability is cumulative:

$$0.75 = 1 - e^{-5\lambda_A} \implies e^{-5\lambda_A} = 0.25 \implies \lambda_A = -\frac{\ln(0.25)}{5} = \frac{\ln 4}{5} \approx 0.277 \text{ payments/min}.$$

For B, the given probability is a survival probability:

$$0.50 = e^{-5\lambda_B} \implies \lambda_B = -\frac{\ln(0.50)}{5} = \frac{\ln 2}{5} \approx 0.139 \text{ payments/min}.$$

Define $N_A(t), N_B(t)$ as the payment counts from the common observation time. For a constant-rate Poisson process,

$$E[N_A(t)] = \lambda_A t, \qquad E[N_B(t)] = \lambda_B t.$$

Thus $E[N_A(t)] - E[N_B(t)] = (\lambda_A - \lambda_B)t$.

$$(\lambda_A - \lambda_B)t = 3 \implies t = \frac{3}{\ln 4/5 - \ln 2/5} = \boxed{\frac{15}{\ln 2} \approx 21.640 \text{ min}}.$$

This is the time when the *difference in expectations* is 3; the actual random count difference need not equal 3 at that time.

Return to this task

8 Two labelling stations [C5-2]

At a food cooperative, let A and B be labelling times in minutes at two stations. Both have support $[3, 15]$.

For A, the density increases along one straight line from zero at 3 to its highest value at 15. For B, equal-length time intervals within the support have equal probabilities. Both densities are zero outside $[3, 15]$.

Questions/Tasks

C5 Identify A's distribution and calculate its mean, variance, and standard deviation. **[1]**

C5 Identify B's distribution and calculate its mean, variance, and standard deviation. **[1]**

Solution: Increasing-linear labelling time

The straight increasing density is an endpoint triangular model: $A \sim \text{Tri}(3, 15, 15)$, support $[3, 15]$ min. For an increasing-linear density on $[a, b]$,

$$\mu = \frac{a + 2b}{3}, \qquad \sigma^2 = \frac{(b-a)^2}{18}, \qquad \sigma = \sqrt{\sigma^2}.$$

$$E(A) = \frac{3 + 2(15)}{3} = 11 \text{ min}, \qquad \text{Var}(A) = \frac{(15-3)^2}{18} = 8 \text{ min}^2, \qquad \sigma_A = \sqrt{8} \approx 2.828 \text{ min}.$$

Return to this task

Solution: Uniform labelling time

Equal-length intervals have equal probabilities, so $B \sim U(3, 15)$, support $[3, 15]$ min. The formulas are $\mu = (a + b)/2$, $\sigma^2 = (b - a)^2/12$, $\sigma = \sqrt{\sigma^2}$.

$$E(B) = \frac{3 + 15}{2} = 9 \text{ min}, \qquad \text{Var}(B) = \frac{(15-3)^2}{12} = 12 \text{ min}^2, \qquad \sigma_B = \sqrt{12} \approx 3.464 \text{ min}.$$

Return to this task

9 Display production and daily revenue [C5-2]

A workshop models display production time D on $[1, 10]$ hours. Its density decreases along one straight line from its highest value at 1 to zero at 10 and is zero outside this support.

The workshop models daily revenue R, measured in **thousands of Colombian pesos (COP)**, by an exact symmetric bell curve on the real line. Its density is centred at 1800 and has inflection points at 1680 and 1920, all in thousands of COP.

Questions/Tasks

C5 Identify D's distribution and calculate its mean, variance, and standard deviation. **[1]**

C5 Identify R's distribution and infer its parameters. Calculate the mean and standard deviation in **COP** and the variance in **COP**^{2}. Interpret the mean in context. **[1]**

Solution: Decreasing-linear production time

The straight decreasing density is $D \sim \text{Tri}(1, 10, 1)$, support $[1, 10]$ h. For a decreasing-linear density, $\mu = (2a + b)/3$, $\sigma^2 = (b - a)^2/18$, $\sigma = \sqrt{\sigma^2}$.

$$E(D) = \frac{2(1) + 10}{3} = 4 \text{ h}, \qquad \text{Var}(D) = \frac{(10 - 1)^2}{18} = \frac{9}{2} = 4.500 \text{ h}^2, \qquad \sigma_D = \frac{3}{\sqrt{2}} \approx 2.121 \text{ h}.$$

Return to this task

Solution: Normal revenue and conversion to COP

The exact symmetric bell curve is normal. In thousands of COP, its centre gives $\mu_R = 1800$. A normal density's inflection points are $\mu_R \pm \sigma_R$, hence

$$\sigma_R = 1920 - 1800 = 1800 - 1680 = 120, \qquad R \sim N(1800, 120^2) = N(1800, 14400).$$

The mathematical support is $\mathbb{R}$. In these units $E(R) = \mu_R$, $\text{Var}(R) = \sigma_R^2$ and $\text{SD}(R) = \sigma_R$. Let $Y = 1000R$ be revenue in COP. In general,

$$E(cR) = cE(R), \quad \text{Var}(cR) = c^2 \text{Var}(R), \quad \text{SD}(cR) = |c| \text{SD}(R).$$

Therefore

$$E(Y) = 1000(1800) = 1{,}800{,}000 \text{ COP},$$

$$\text{Var}(Y) = 1000^2(14400) = 14{,}400{,}000{,}000 \text{ COP}^2, \qquad \text{SD}(Y) = 1000(120) = 120{,}000 \text{ COP}.$$

The workshop's long-run average daily revenue is 1,800,000 COP; individual days vary around that average.

Return to this task

10 Requests and repair times [C5-2]

A repair cooperative receives requests according to a constant-rate Poisson process averaging 12 requests per hour. Let T be the interarrival time in minutes.

The duration S of a repair has support $[10, 30]$ minutes. Its density rises linearly from zero at 10 to a single peak at 18, then falls linearly to zero at 30; it is zero outside this range.

Questions/Tasks

C5 Identify T's distribution and calculate its mean and standard deviation in minutes and its variance in minutes^2. **[1]**

C5 Identify S's distribution and calculate its mean and standard deviation in minutes and its variance in minutes^2. **[1]**

Solution: Exponential interarrival time

The constant-rate Poisson process gives $T \sim \text{Exp}(\lambda)$, support $[0, \infty)$ min, with $\lambda = 12/60 = 1/5 = 0.200$ requests/min. The formulas are $\mu = 1/\lambda$, $\sigma^2 = 1/\lambda^2$ and $\sigma = 1/\lambda$.

$$E(T) = \frac{1}{1/5} = 5 \text{ min}, \qquad \text{Var}(T) = \frac{1}{(1/5)^2} = 25 \text{ min}^2, \qquad \sigma_T = \frac{1}{1/5} = 5 \text{ min}.$$

Return to this task

Solution: Triangular repair duration

The two straight branches give $S \sim \text{Tri}(10, 30, 18)$, support $[10, 30]$ min. For minimum a, maximum b and mode c,

$$\mu = \frac{a+b+c}{3}, \qquad \sigma^2 = \frac{a^2+b^2+c^2-ab-ac-bc}{18}, \qquad \sigma = \sqrt{\sigma^2}.$$

$$E(S) = \frac{10+30+18}{3} = \frac{58}{3} \approx 19.333 \text{ min}.$$

$$\text{Var}(S) = \frac{10^2+30^2+18^2-10(30)-10(18)-30(18)}{18}$$

$$= \frac{304}{18} = \frac{152}{9} \approx 16.889 \text{ min}^2,$$

$$\sigma_S = \sqrt{152/9} = \frac{2\sqrt{38}}{3} \approx 4.110 \text{ min}.$$

Return to this task

11 Parcel masses [C6-2]

A sample of 50 parcel masses X, in kilograms, is recorded in two non-overlapping parts. The first 20 masses are listed individually:

1.2	1.5	2.0	2.2	2.4
2.5	2.7	3.0	3.1	3.4
1.8	2.0	2.3	2.6	2.9
3.0	3.2	3.5	3.8	4.0

The remaining 30 masses are summarised in a frequency table:

x_i	2.0	2.4	2.6	2.8	3.0	3.2
f_i	3	5	7	6	5	4

The model gives $P(2 \le X < 3) = 0.40$; include 2 and exclude 3. Combine both parts as one sample.

Questions/Tasks

C6 Find the empirical probability of this event. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

In the first part, the eligible masses are $2.0, 2.2, 2.4, 2.5, 2.7, 2.0, 2.3, 2.6, 2.9$. There are **9** eligible raw observations. In the table, values $2.0, 2.4, 2.6, 2.8$ contribute $3 + 5 + 7 + 6 = 21$. Include 2 and exclude every occurrence of 3. The parts do not overlap, so $n = 20 + 30 = 50$ and $k = 9 + 21 = 30$.

$$\widehat{P}(2 \le X < 3) = \frac{k}{n} = \frac{30}{50} = \frac{3}{5} = 0.6000 = 60.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$ with the exact relative frequency.

$$E = \left|\frac{3}{5} - 0.40\right| = \frac{1}{5} = 0.2000.$$

$$E\% = \frac{\frac{1}{5}}{0.40} \times 100\% \approx 50.00\%.$$

Probabilities and their errors are unitless.

Return to this task

12 File transfers, grouped data [C6-2]

Transfer times X in seconds are grouped below. Classes include the left endpoint and exclude the right endpoint.

Class	$[0,4)$	$[4,8)$	$[8,12)$	$[12,16)$
f_i	3	9	5	3

The model gives $P(4 \le X < 12) = 0.75$.

Questions/Tasks

C6 Find the empirical probability of this event. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

The complete included classes are $[4,8)$ and $[8,12)$. Thus $n = 3+9+5+3 = 20$ and $k = 9+5 = 14$. These are exact event counts; no midpoint approximation is needed.

$$\widehat{P}(4 \le X < 12) = \frac{k}{n} = \frac{14}{20} = \frac{7}{10} = 0.7000 = 70.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$ with the exact relative frequency.

$$E = \left|\frac{7}{10} - 0.75\right| = \frac{1}{20} = 0.0500.$$

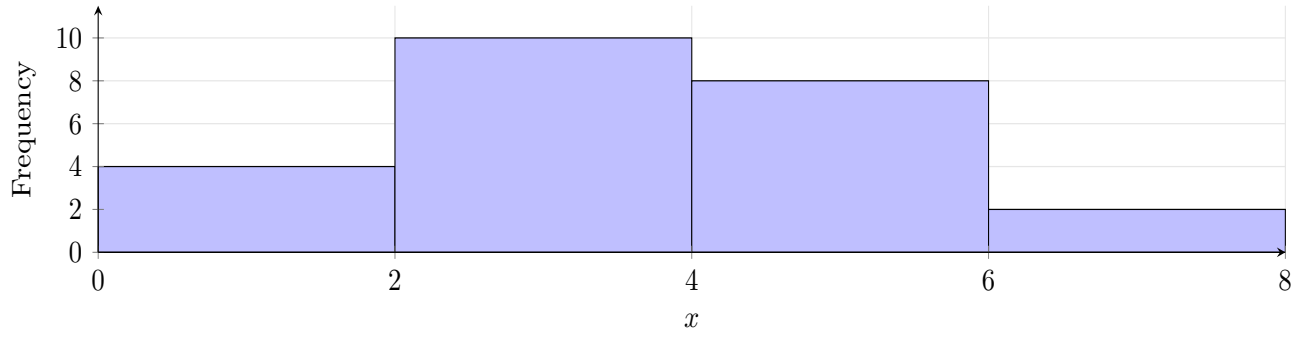
$$E\% = \frac{\frac{1}{20}}{0.75} \times 100\% \approx 6.67\%.$$

Probabilities and their errors are unitless.

Return to this task

13 Collection queues, histogram [C6-2]

The histogram is the only supplied sample record of collection waiting times X in minutes. Equal-width classes include the left endpoint and exclude the right endpoint. Bar heights are frequencies; the horizontal axis gives minutes.



The model gives $P(2 \le X < 6) = 0.70$.

Questions/Tasks

C6 Find the empirical probability of this event. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

Read histogram heights as frequencies: $4, 10, 8, 2$. The two complete included classes are $[2, 4)$ and $[4, 6)$. Thus $n = 4 + 10 + 8 + 2 = 24$ and $k = 10 + 8 = 18$. No midpoint approximation is needed.

$$\widehat{P}(2 \le X < 6) = \frac{k}{n} = \frac{18}{24} = \frac{3}{4} = 0.7500 = 75.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$ with the exact relative frequency.

$$E = \left|\frac{3}{4} - 0.70\right| = \frac{1}{20} = 0.0500.$$

$$E\% = \frac{\frac{1}{20}}{0.70} \times 100\% \approx 7.14\%.$$

Probabilities and their errors are unitless.

Return to this task

14 Cold-room readings, mixed data [C7-2]

A sample of 12 cold-room readings, in degrees Celsius, is recorded in two non-overlapping parts. The first five readings are $6, 8, 9, 11, 12$. The remaining seven readings are summarised below:

x_i	6	9	12
f_i	2	3	2

Combine both parts as one sample. The theoretical mean is $9\,^\circ\text{C}$ and variance is $5\,(^\circ\text{C})^2$.

Questions/Tasks

C7 Calculate mean, variance and standard deviation. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from combined data

The first five readings contribute one each at $6, 8, 9, 11, 12$; the frequency table adds $2, 3, 2$ at $6, 9, 12$. The combined sample has the following actual values and frequencies, so no midpoint estimation is used.

x_i	6	8	9	11	12
f_i	3	1	4	1	3

Use the sample-variance denominator $n-1$, not n:

$$n = \sum f_i = 12, \qquad \bar{x} = \frac{\sum f_i x_i}{n} = \frac{3(6)+1(8)+4(9)+1(11)+3(12)}{12} = \frac{109}{12} \approx 9.083\ ^\circ\text{C}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (x_i - \bar{x})^2 \\ &= 3\left(6 - \frac{109}{12}\right)^2 + 1\left(8 - \frac{109}{12}\right)^2 + 4\left(9 - \frac{109}{12}\right)^2 + 1\left(11 - \frac{109}{12}\right)^2 + 3\left(12 - \frac{109}{12}\right)^2 \\ &= \frac{707}{12}\ (^\circ\text{C})^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{n-1} = \frac{\frac{707}{12}}{12-1} = \frac{707}{132} \approx 5.356\ (^\circ\text{C})^2,$$

$$s = \sqrt{s^2} = \sqrt{\frac{707}{132}} \approx 2.314\ ^\circ\text{C}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$ and $E\% = 100E/|z_0|$. Using unrounded statistics:

$$E_{\bar{x}} = \left|\frac{109}{12} - 9\right| = \frac{1}{12} \approx 0.083\ ^\circ\text{C}, \qquad E_{\bar{x}}\% = 100\frac{\frac{1}{12}}{9}\% \approx 0.93\%.$$

$$E_{s^2} = \left|\frac{707}{132} - 5\right| = \frac{47}{132} \approx 0.356\ (^\circ\text{C})^2, \qquad E_{s^2}\% = 100\frac{\frac{47}{132}}{5}\% \approx 7.12\%.$$

The requested errors concern the mean and variance only.

Return to this task

15 Courier routes, grouped data [C7-2]

Route times in minutes are grouped below. Classes include the left endpoint and exclude the right endpoint. The theoretical mean is 14 minutes and variance is 12 minutes^2.

Class	$[8, 12)$	$[12, 16)$	$[16, 20)$
f_i	2	3	1

Questions/Tasks

C7 Calculate mean, variance and standard deviation. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from class midpoints

The class midpoint is (left + right)/2. Individual observations are unavailable, so all three statistics and their errors below are **midpoint estimates**.

m_i	10	14	18
f_i	2	3	1

Use the sample-variance denominator $n - 1$, not n:

$$n = \sum f_i = 6, \qquad \bar{x} = \frac{\sum f_i m_i}{n} \approx \frac{2(10) + 3(14) + 1(18)}{6} = \frac{40}{3} \approx 13.333 \text{ min}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i(m_i - \bar{x})^2 \\ &= 2\left(10 - \frac{40}{3}\right)^2 + 3\left(14 - \frac{40}{3}\right)^2 + 1\left(18 - \frac{40}{3}\right)^2 \\ &= \frac{136}{3}\ \text{min}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i(m_i - \bar{x})^2}{n-1} \approx \frac{\frac{136}{3}}{6-1} = \frac{136}{15} \approx 9.067 \text{ min}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{136}{15}} \approx 3.011 \text{ min}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$ and $E\% = 100E/|z_0|$. Using unrounded midpoint estimates:

$$E_{\bar{x}} = \left|\frac{40}{3} - 14\right| = \frac{2}{3} \approx 0.667 \text{ min}, \qquad E_{\bar{x}}\% = 100\frac{\frac{2}{3}}{14}\% \approx 4.76\%.$$

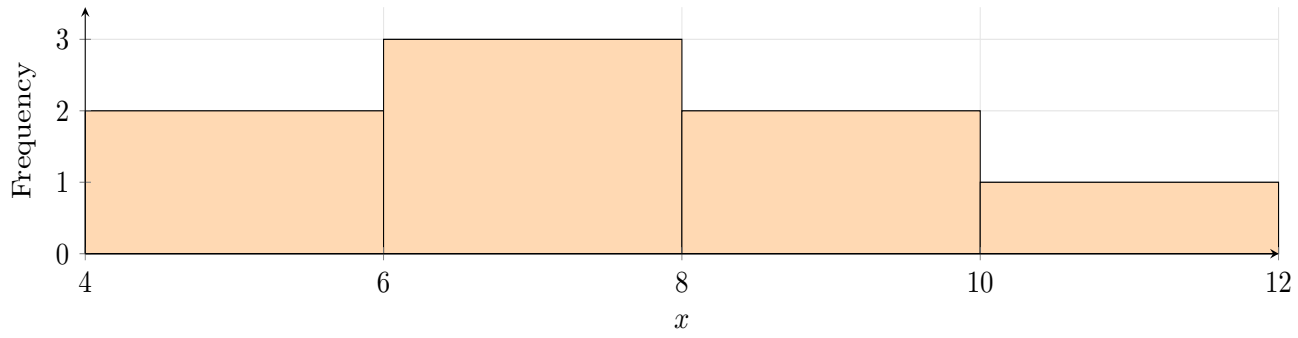
$$E_{s^2} = \left|\frac{136}{15} - 12\right| = \frac{44}{15} \approx 2.933 \text{ min}^2, \qquad E_{s^2}\% = 100\frac{\frac{44}{15}}{12}\% \approx 24.44\%.$$

The requested errors concern the mean and variance only.

Return to this task

16 Reusable-container masses [C7-2]

The histogram is the only supplied sample record of container masses in kilograms. Equal-width classes include the left endpoint and exclude the right endpoint. Bar heights are frequencies; the horizontal axis gives kilograms. The theoretical mean is 7 kg and variance is 4 kg^2.



Questions/Tasks

C7 Calculate mean, variance and standard deviation. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from class midpoints

The class midpoint is (left + right)/2. Individual observations are unavailable, so all three statistics and their errors below are **midpoint estimates**.

m_i	5	7	9	11
f_i	2	3	2	1

Use the sample-variance denominator $n-1$, not n:

$$n = \sum f_i = 8, \qquad \bar{x} = \frac{\sum f_i m_i}{n} \approx \frac{2(5)+3(7)+2(9)+1(11)}{8} = \frac{15}{2} \approx 7.500 \text{ kg}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (m_i - \bar{x})^2 \\ &= 2\left(5 - \frac{15}{2}\right)^2 + 3\left(7 - \frac{15}{2}\right)^2 + 2\left(9 - \frac{15}{2}\right)^2 + 1\left(11 - \frac{15}{2}\right)^2 \\ &= 30 \text{ kg}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1} \approx \frac{30}{8-1} = \frac{30}{7} \approx 4.286 \text{ kg}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{30}{7}} \approx 2.070 \text{ kg}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$ and $E\% = 100E/|z_0|$. Using unrounded midpoint estimates:

$$E_{\bar{x}} = \left|\frac{15}{2} - 7\right| = \frac{1}{2} \approx 0.500 \text{ kg}, \qquad E_{\bar{x}}\% = 100\frac{\frac{1}{2}}{7}\% \approx 7.14\%.$$

$$E_{s^2} = \left|\frac{30}{7} - 4\right| = \frac{2}{7} \approx 0.286 \text{ kg}^2, \qquad E_{s^2}\% = 100\frac{\frac{2}{7}}{4}\% \approx 7.14\%.$$

The requested errors concern the mean and variance only.

Return to this task

17 Summarized Solutions

Exact parameters and counts are retained. See the linked problems for procedures and supports.

C1 – All twelve variables

Problem 1. D = discrete; C = continuous.

Context	Variable	Class	Model (parameters as defined above)
1	a	D	Bernoulli(0.92)
1	b	D	Binomial$(40, 0.92)$
2	a	D	Geometric(0.08), trials including the first incorrect invoice
2	b	C	$U(-3, 3)$ g
3	a	D	Poisson(2) orders per hour
3	b	C	Tri$(10, 30, 18)$ min
4	a	C	Increasing-linear: $f(x) = 2x$ on $[0, 1]$
4	b	C	$N(0, 16)$, in grams
5	a	D	Binomial$(50, 0.04)$
5	b	D	Poisson(15) enquiries over three days
6	a	C	Candidate Tri$(15, 45, 15)$ min if linearity is assumed
6	b	C	Candidate $N(5, 1.44)$, in kg, if normality is assumed

Context 6 is not uniquely determined: monotone decrease need not be linear, and symmetry with SD 1.2 kg need not be normal. The normal candidate has support $\mathbb{R}$; physical mass is nonnegative and its precise support is not supplied.

C2 – Triangular models

Problem 2. Both models have mode 13 min and are zero outside their respective supports.

Support/model	Rising pdf	Falling pdf	$P(9 \le S \le 19)$
$[5, 25]$, Tri$(5, 25, 13)$	$(s-5)/80$	$(25-s)/120$	$3/4 = 0.7500 = 75.00\%$
$[0, 30]$, Tri$(0, 30, 13)$	$s/195$	$(30-s)/255$	$368/663 \approx 0.5551 = 55.51\%$

Rising branches end at 13; falling branches start after 13. Heights are $1/10$ and $1/15$ min^{-1}.

C3 – Cycles and directions

Problem 3: restart window $[52, 60) \cup [0, 11]$ has probability $19/60 \approx 0.3167 = 31.67\%$. The 200-second tolerance on each side gives $[0, 10/3] \cup [170/3, 60)$ and probability $1/9 \approx 0.1111 = 11.11\%$.

Problem 4: $P_{\pi/9} = 1/9 \approx 0.1111 = 11.11\%$; $P_{15^\circ} = 1/12 \approx 0.0833 = 8.33\%$. The former exceeds the latter by $1/36 \approx 0.0278$, or 2.78 percentage points (relative increase 33.33%).

C4 – Waiting probabilities and inferred rates

Problem 5. Rates below are openings/minute.

Scenario	Rate	Exact probability	Decimal and percentage
1	1/6	$P(T < 4) = 1 - e^{-2/3}$	0.4866; 48.66%
2	1/4	$P(T > 6) = e^{-3/2}$	0.2231; 22.31%
3	3/20	$P(2 < T < 7) = e^{-3/10} - e^{-21/20}$	0.3909; 39.09%
4	1/5	$P(T < 3 \text{ or } T > 8) = 1 - e^{-3/5} + e^{-8/5}$	0.6531; 65.31%

Problem 6: morning $\lambda_m = -\ln(0.40)/10 \approx 0.092$ receipts/min, $60\lambda_m \approx 5.498$ receipts/hour. Afternoon $\lambda_a = -\ln(0.20)/6 \approx 0.268$ receipts/min, $60\lambda_a \approx 16.094$ receipts/hour.

Problem 7: $\lambda_A = \ln 4/5 \approx 0.277$ and $\lambda_B = \ln 2/5 \approx 0.139$ payments/min. The expected-count difference is 3 at $t = 15/\ln 2 \approx 21.640$ min.

C5 – Theoretical moments

Supports: $A, B \in [3, 15]$ min; $D \in [1, 10]$ h; revenue's normal support $\mathbb{R}$; $T \in [0, \infty)$ min; repair $S \in [10, 30]$ min.

Problem	Variable/model	Mean	Variance	SD	Units
8	$A \sim \text{Tri}(3, 15, 15)$	11	8	2.828	min; min^2
8	$B \sim U(3, 15)$	9	12	3.464	min; min^2
9	$D \sim \text{Tri}(1, 10, 1)$	4	4.500	2.121	h; h^2
10	$T \sim \text{Exp}(1/5)$	5	25	5	min; min^2
10	$S \sim \text{Tri}(10, 30, 18)$	19.333	16.889	4.110	min; min^2

Units are first for mean and SD, second for variance. Exact repair moments: 58/3, 152/9 and $2\sqrt{38}/3$. Problem 9, revenue: $R \sim N(1800, 14400)$ in thousands of COP, with SD 120. In COP: mean 1,800,000, variance 14,400,000,000 COP^2, SD 120,000 COP. The mean is the long-run average daily revenue.

C6 – Empirical probabilities

$E = |\widehat{P} - P_0|$, $E\% = 100E/|P_0|$. Counts use complete event classes, not midpoints.

Problem	Event	n	k	$\widehat{P}$	P_0	E	$E\%$
11	$2 \le X < 3$ kg	50	30	0.6000	0.4000	0.2000	50.00%
12	$4 \le X < 12$ s	20	14	0.7000	0.7500	0.0500	6.67%
13	$2 \le X < 6$ min	24	18	0.7500	0.7000	0.0500	7.14%

Parcel counts: 9 eligible raw masses plus 21 table observations, including 2 and excluding 3.

C7 – Sample statistics and requested errors

All sample variances use $n-1$. Asterisks mark midpoint estimates because individual observations are unavailable.

Problem	n	$\bar{x}$	s^2	s	Units
14	12	9.083	5.356	2.314	°C; $(°C)^2$
15*	6	13.333	9.067	3.011	min; min^2
16*	8	7.500	4.286	2.070	kg; kg^2

Exact statistics $(\bar{x}, s^2)$: $(109/12, 707/132)$, $(40/3, 136/15)$, $(15/2, 30/7)$.

Problem	Statistic	Absolute error	Percentage error	Absolute-error units
14	Mean	0.083	0.93%	°C
	Variance	0.356	7.12%	$(°C)^2$
15*	Mean	0.667	4.76%	min
	Variance	2.933	24.44%	min^2
16*	Mean	0.500	7.14%	kg
	Variance	0.286	7.14%	kg^2

